ITCS216: Fundamentals of Database Systems (1/2026)

Faculty of Information and Communication Technology

Project Phase II: Database Implementation

(Total: 15% of Grade)

Description:

In this project, students will act as *database administrators* of the project who are responsible for converting the design based on the relational data model and data dictionary from the previous phase to create the real database using relational database management (e.g., MySQL Server). Finally, all design and implementation must be documented in the final report.

Members: 5 Members (same as the previous group); continued from Phase I

Requirements and Deliverables

There are **2** milestones across Phase II and Document submission; each comes with the deliverables:

Milestone 1 (M1): ERD/EERD Transformation

In M1, students must create the relational data models based on 8-step transformations (**Hint**: Lectures on Relational Data Model) as follows:

1. **Revise** the Entity Relationship Diagram/Extended ERD in Project Phase I.
2. **Select 8-10** related entities from the revised ERD/EERD in Step 1). The ERD must have
   - ☐ The selection contains 8 to 10 entities.
   - ☐ All selected entities connect to the same model.
   - ☐ The selection includes one supertype and two recommended subtypes.
   - ☐ The selection includes at least one weak entities.
   - ☐ The selection includes at least one multivalued attribute
   - ☐ The selection includes at least one M to N relationship.
3. **Discuss** with your instructor for the appropriate entities.
4. **Transform** the selected ERD into a relational schema by following the 8-step transformation. Show step-by-step clearly.
5. **Draw** the final relational schema.

Deliverables: A short report consisting of the revised and selected ERD/EERD, Step-by-step transformation, and the final relational schema, named your report "***PJ2M1-SecXXGRXX.pdf***"

Submission: Saturday, October 31, 2026, at 11:59 PM

Milestone 2 (M2): Database Implementation and Query

In M2, students must implement the database of the selected business domain from the designed schema. There are 3 tasks to implement including

Task 1: Data Dictionary, DDL, and DML

In this task, students must create a data dictionary and write the SQL (DDL and DML) script to create the database with the data records using MySQL Server.

1. From the relational schema from M1, **create** a **data dictionary** consisting of all relations.
   The details of a data dictionary must at least include:

Table Name	Attribute Name	Contents	Type	Format	Nullable	Range	Key	FK Reference Table

2. **Create** a database, and name the database "`secXXgrXXdb`", using Data Definition Language (DDL), e.g., `CREATE`, `ALTER`, and `DROP` commands *if needed*.
3. **Create** tables and constraints using the DDL commands designed in Step 1).
4. **Insert** data records into each table using realistic data with the Data Manipulation Language (DML), e.g., `INSERT`, `UPDATE`, and `DELETE` *if needed*, under the following conditions:
   - The ***Master Table*** must have at least 10 rows of data.
   - The ***Transaction Table*** must have at least 30 rows of data.
5. **Save** your SQL commands from Step 2) to Step 4) into one file, named "`pj2secXXgrXXdb.sql`"

Note:

- The database must conform to the Relational Data Model and Data Dictionary.
- Use the DDL commands (`CREATE`, `ALTER`, and `DROP` commands) as needed. Once running, the SQL script must create the new database with the data records.
- The master table stores basic business information that rarely changes, such as product and customer data. In contrast, the transaction table stores actions or events involving these entities, like purchases or enrollments, and this data frequently updates.
- The realistic data means the data are closely related to the real world according to your business domain. DO NOT use unrealistic data, e.g., the customer's name is "aaaa" or the product's name is "bbbb".

Output of Task 1: Data dictionary and the SQL script named "`pj2secXXgrXXdb.sql`"

Task 2: Database Query

In this task, students are required to **produce 5 queries per member** that are meaningful to your business domain using Data Query Language.

1. Each member **creates** 5 "meaningful" SQL queries with the minimum requirement of each query as follows:
   - **SQL Query 1:** Must have a `WHERE` clause with at least two or more conditions.
   - **SQL Query 2:** Must use SQL built-in functions, for example, string function, numerical function, or date function.
   - **SQL Query 3:** Must use aggregate functions, for example, `SUM`, `AVG`, `MIN`, `MAX`, with `GROUP BY` and `HAVING`
   - **SQL Query 4:** Must use `INNER JOIN`
   - **SQL Query 5:** Must use any type of `OUTER JOIN`

2. Save all SQL commands from Step 1) of each member into one file, named "pj2secXXgrXXquery.sql"

Note:

- Each member must have a different "structure" of the query, i.e., query from different tables. Minor changes in the values of the condition, the attributes, and the functions are considered the same SQL structure.
- Each set of commands should have a clear purpose for retrieving data, and the displayed data should be beneficial to the respective business.

Output of Task 2: the SQL script named "pj2secXXgrXXquery.sql"

Task 3: Final Report

The Final Report on Project Phase 2 in PDF format, including:

- Your brief description of the business domain (from your Phase I)
- Everything about Relational Schema from **M1** including your selected ERD and the 8-step transformation to the Relational schema.
- Data dictionary (Task 1).
- Captured-screen result of your database and each table along with its data from MySQL Server (Task 1)
- Description of each query and its result (Task 2) using the following template:

Template for Query Description

Query #	The List of Students Registered in Class with Code 20040
Owner	<Student ID firstname lastname>
Objective	To display information on students registered in class with code 20040, including student ID and full name.
Query	SELECT s.stu_num, CONCAT(s.stu_fname,' ', s.stu_lname) AS stu_name FROM Student s JOIN Enroll e ON s.stu_num = e.stu_num JOIN Class cl ON e.class_code = cl.class_code WHERE cl.class_code = '20040';
Result	stu_num | stu_name 324269 | Walter Oblonski 328993 | Anton Jubilar

Note: The owner is the information of the student who wrote this query and for the result, please ensure the clarity of screen capture images in the report.

- AI Usage Acknowledgment Statement (if applicable) *See the details below*
- References

Deliverables:

The final report named **PJ2-secXXgrXX.pdf** and two SQL files named:

- pj2secXXgrXXdb.sql from Task 1
- pj2secXXgrXXquery.sql from Task 2

Note: The .sql file must be runnable on any machine with MySQL Server. When executed, it should create the database and tables with data. Please check the syntax before submission.

Due Date: Sunday, November 29, 2026, at 11:59 PM

Acknowledgement Usage of AI for Project

To support your learning, Generative AI tools may be used only to support tasks such as data generation or code debugging. Grammarly may be used exclusively for editing and refining project reports.

If any Generative AI tools are used, students must include an AI Usage Acknowledgment Statement in their report as specified.

AI Usage Acknowledgment Statement

We acknowledge that we have utilized Generative Artificial Intelligence (Generative AI) tools (_____,____,_____) in preparing this work. The AI tools were used for the following purposes:

☐ Brainstorming ideas

☐ Data generation

☐ Language correction, grammar checking, or translation

☐ Code editing and debugging

☐ Other (please specify): _____

Rubric for Phase II:

Topic	5 Excellent	4 Good	3 Moderate	2 Fair	1 Poor	0 N/A
Schema	Level 3 and the diagram incorporates the results from each step and connect into a complete relational model	Level 3 and the diagram does not incorporate the results from or missing more than 2 steps	All 8 steps are clearly shown included "Not applicable"	Some steps are clearly shown but some steps are skipped without stating the reasons or no final relational schema	Not showing step by step process	Do not have schema
Data Dictionary	Level 3 and correct data type and constraints	Level 3 but incorrect data type and constraints	Level 1 and Correctly conform to the design schema	Level 1 and Partially conform to the design schema	Have data dictionary detail as required	Do not have a data dictionary
DDL and DML	The script generates the full database with the data and align with the design	The script generates the full database with the data and partially align with the design	The script generates a complete database with the data	The script is buggy but contain the DDL to generate a complete database without data	The script is buggy but contain the DDL to generate partial database	Do not have any DDL scripts
Data quality	All tables have **real-world** data as required, i.e., Master > 10 records **and** Trans > 30 records	All table have **real-world** data, partially satisfied the requirement, i.e., Master > 10 records **or** Trans > 30 records	All tables have **real-world** data, somewhat satisfied the requirement, i.e., Master < 10 records **and** Trans < 30 records	Few tables have **unrealistic** data,	Most tables have some **unrealistic** data	All tables do not have data or have unrealistic data
Data query	100% of SQL queries met the minimum requirement,	At least 80% of SQL queries met the minimum requirement,	At least 80% of SQL queries met the minimum	At least 60% of SQL queries met the minimum requirement **and**	Less that 60% of SQL queries met the minimum requirement **and**	All SQL queries does not met the minimum

Topic	5 Excellent	4 Good	3 Moderate	2 Fair	1 Poor	0 N/A
	produced the complete result, **and** has the meaningful objectives	produced the result, **and** has the meaningful objectives	requirement **and** produced the result with partially meaningful objectives	produced the result with partially meaningful objectives	produced the result with partially meaningful objectives	requirement **or** cannot produce the result **or** do not have meaningful objectives
SQL files	**Both** files can be run **all at once** without errors and produced the complete result	**Only one file** from the two can be run **all at once** without errors and produced the complete result	**Both** files can produce the complete result **but** need to run block by block.	**Only one file** can produce the complete result **and** need to run block by block.	**Both** files cannot produce the complete result **and** need to run block by block.	No SQL files are submitted.
Document	All sections are included as required with the good quality	All sections are included but partially missing in detailed explanation or results	All sections are included but partially missing in detailed explanation and no result	Some sections are missing but have details and results	Some sections are missing	No document submission

Penalties for late submission

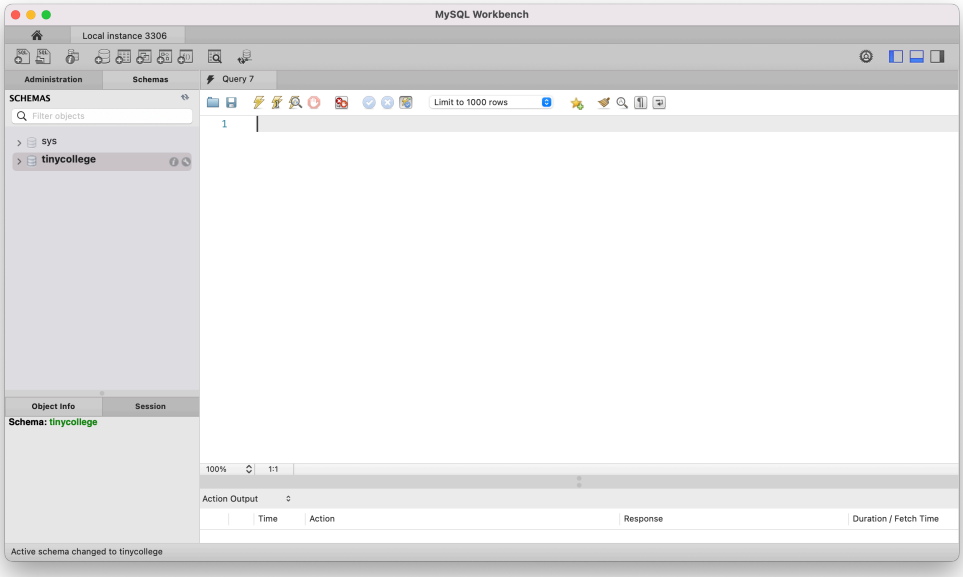
If you hand in the work late, *without a good reason*, your marks will be reduced as follows:

a. A deduction of 10% of the maximum mark available from the actual mark achieved by you will be imposed upon expiry of the deadline. For example, 10% of 100 marks is 10 marks. If the deadline is 5pm and you submit at 6pm on the same day, your mark of 68 would become a mark of 58 after the penalty.

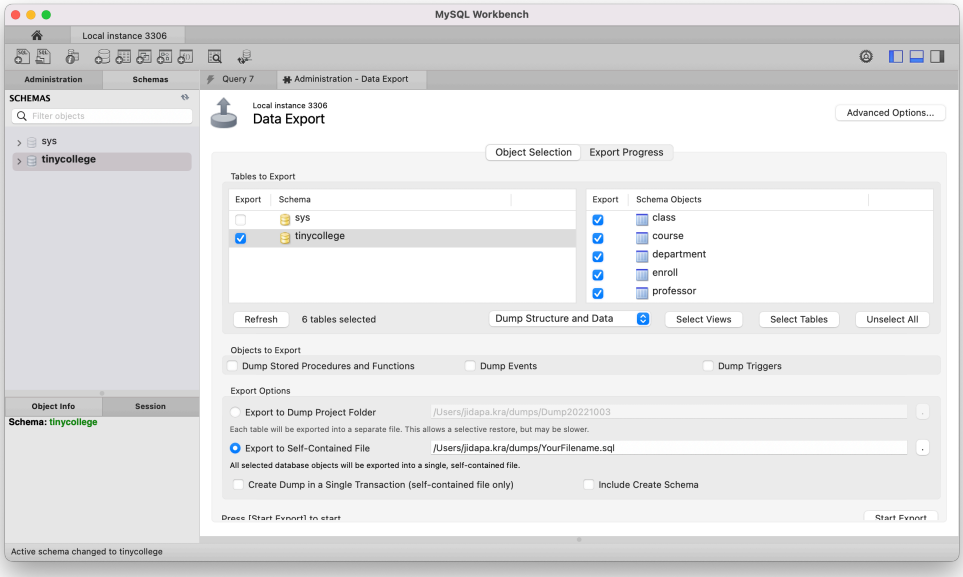
b. A further deduction of 5% of the maximum mark available from the actual mark achieved by you will be imposed on each of the next subsequent days

Appendix: How to export a SQL script

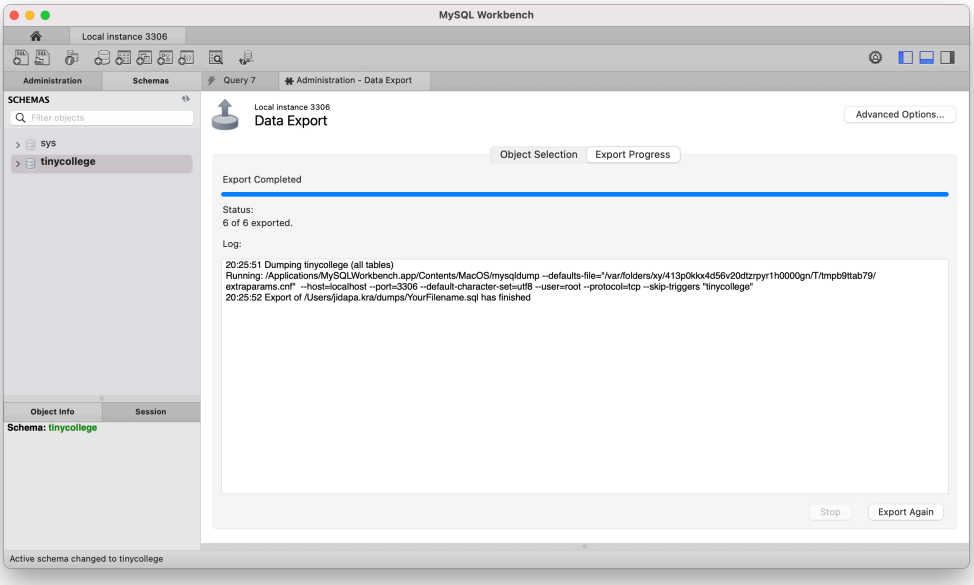
1. Open MySQL Workbench. Make sure you already have the database to backup/export (e.g. tinycollege)



2. On the top menu, select “Server” > “Data Export”
On the “Data Export” window, select the database and schema to be exported. Then, select “**Export to Self-Contained File**” and set the file name. **Note**: your output file will be in *.sql
Click “Start Export”



When finished, you will see the following screen.



You can check your SQL file by re-opening it in the Workbench

